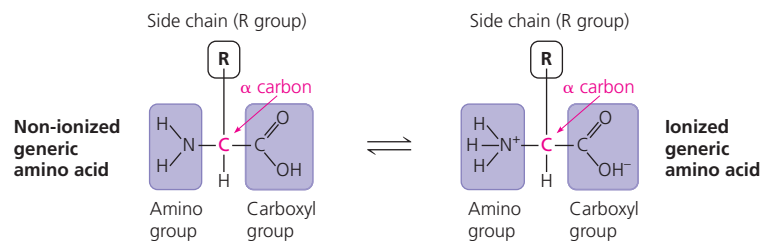
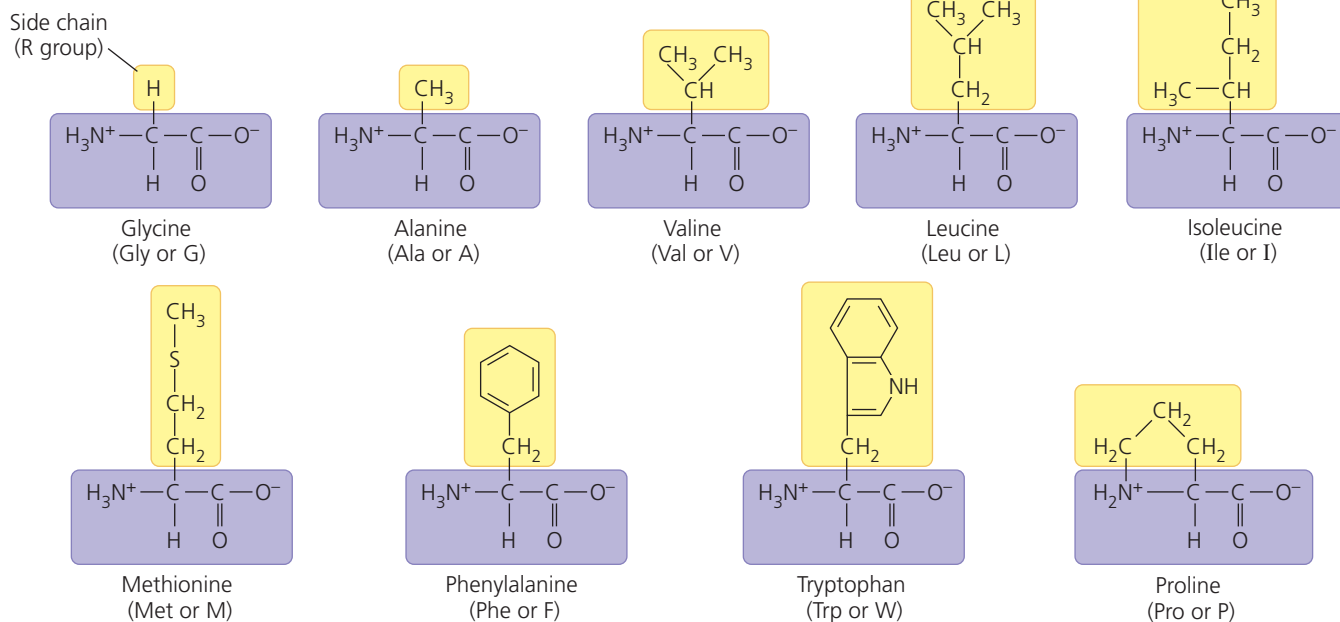


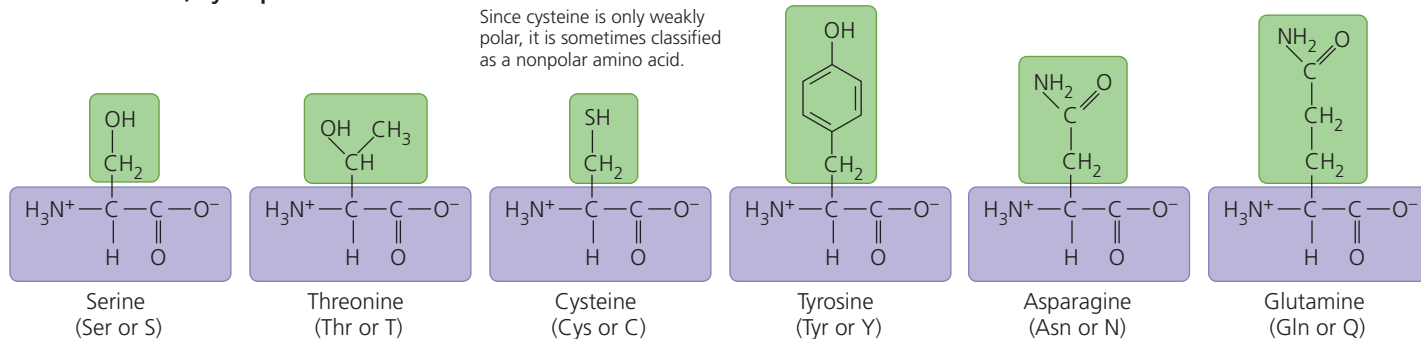
▼ **Figure 5.14 The 20 amino acids of proteins.** At the top right are the non-ionized and ionized forms of a generic amino acid. The specific amino acids are shown below in their ionized form, prevalent at the pH within a cell (pH 7.2). The amino acids are grouped by properties of their side chains. The three-letter and one-letter abbreviations for the amino acids are in parentheses. All of the amino acids used in proteins are L enantiomers (see Figure 4.7c).



Nonpolar side chains; hydrophobic



Polar side chains; hydrophilic



Electrically charged side chains; hydrophilic

